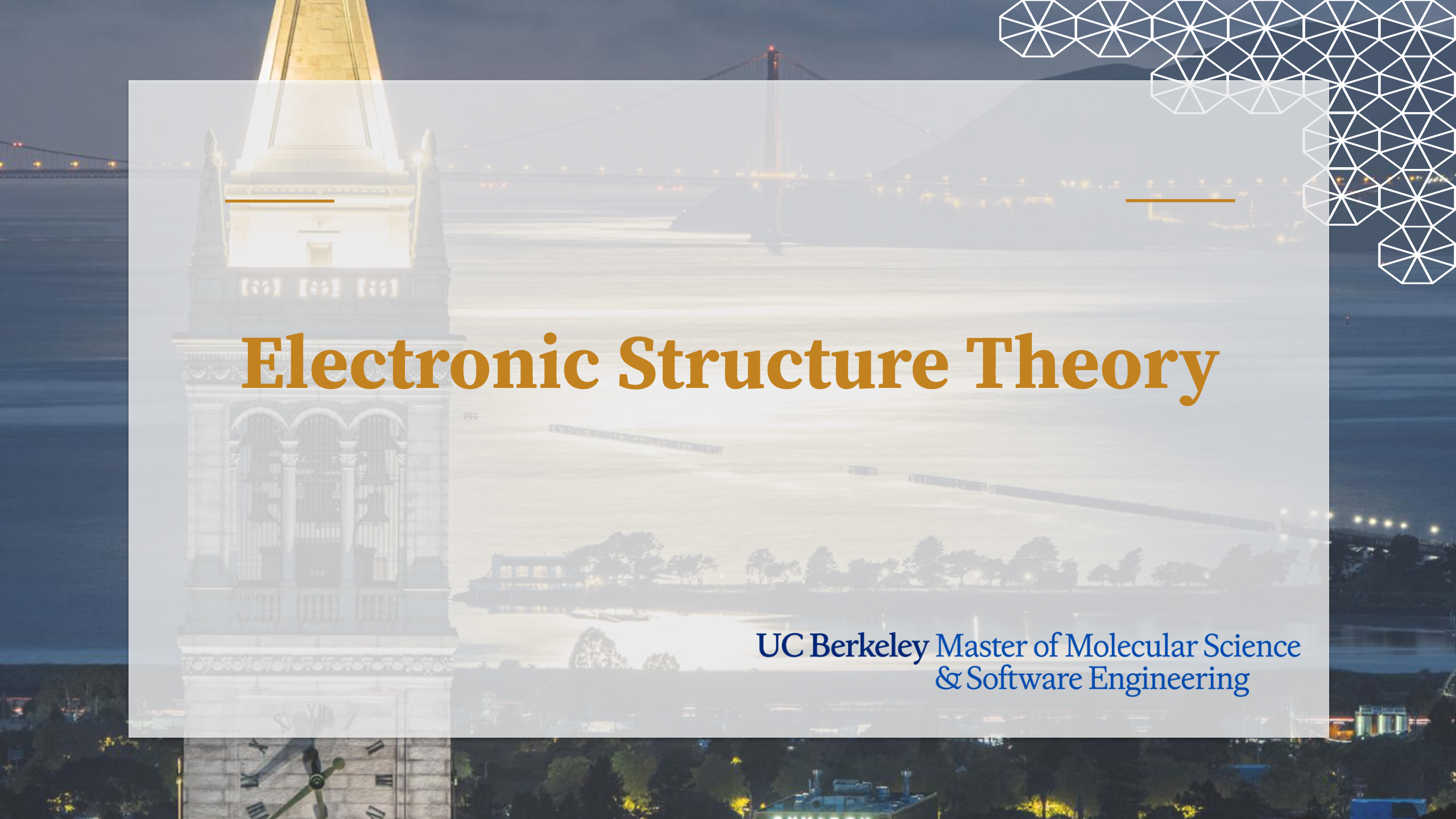




The background of the slide features a night-time photograph of the UC Berkeley Campanile tower on the left, with its clock face visible at the bottom. In the background, the San Francisco Bay Bridge is illuminated against the dark sky, with the city lights of San Francisco visible in the distance. The water of the bay is dark, with a few small boats visible. The overall scene is a blend of architectural and natural elements, representing the university's location.

Electronic Structure Theory

UC Berkeley Master of Molecular Science
& Software Engineering

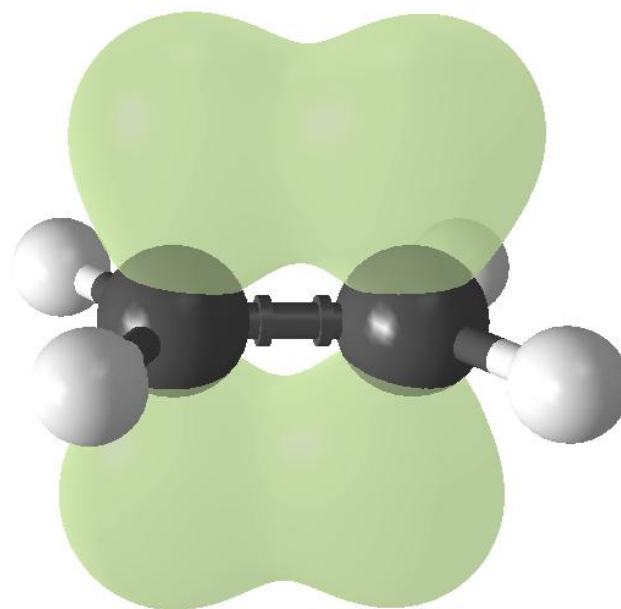
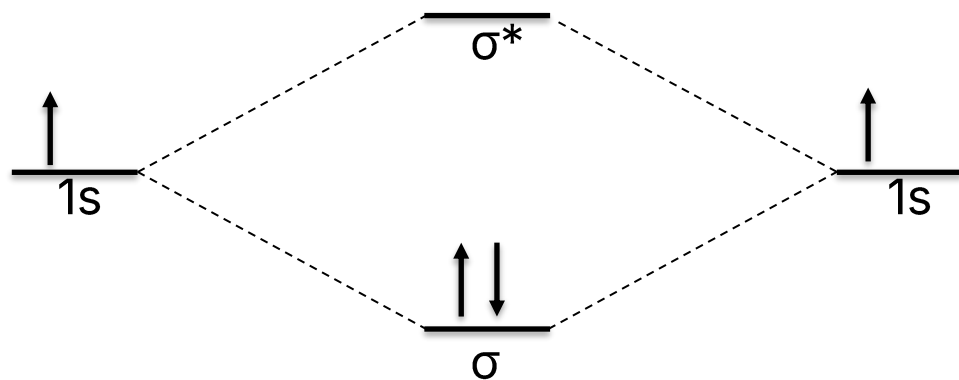
“Scientific knowledge is as much an understanding of the diversity of situations for which a theory or its models are relevant as an understanding of its limits”

– Elinor Ostrom, *Governing the Commons*



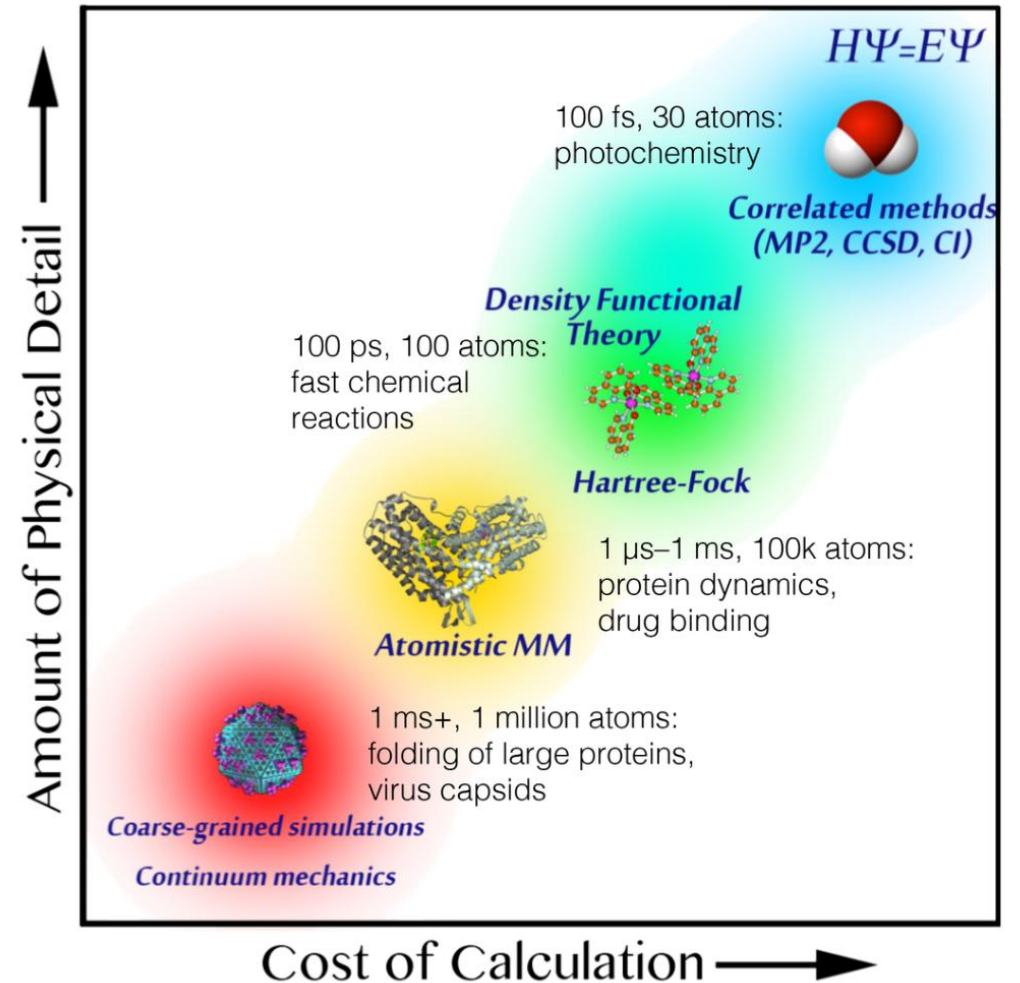
What is Electronic Structure Theory

- Determination of location and structure of electrons within a molecule

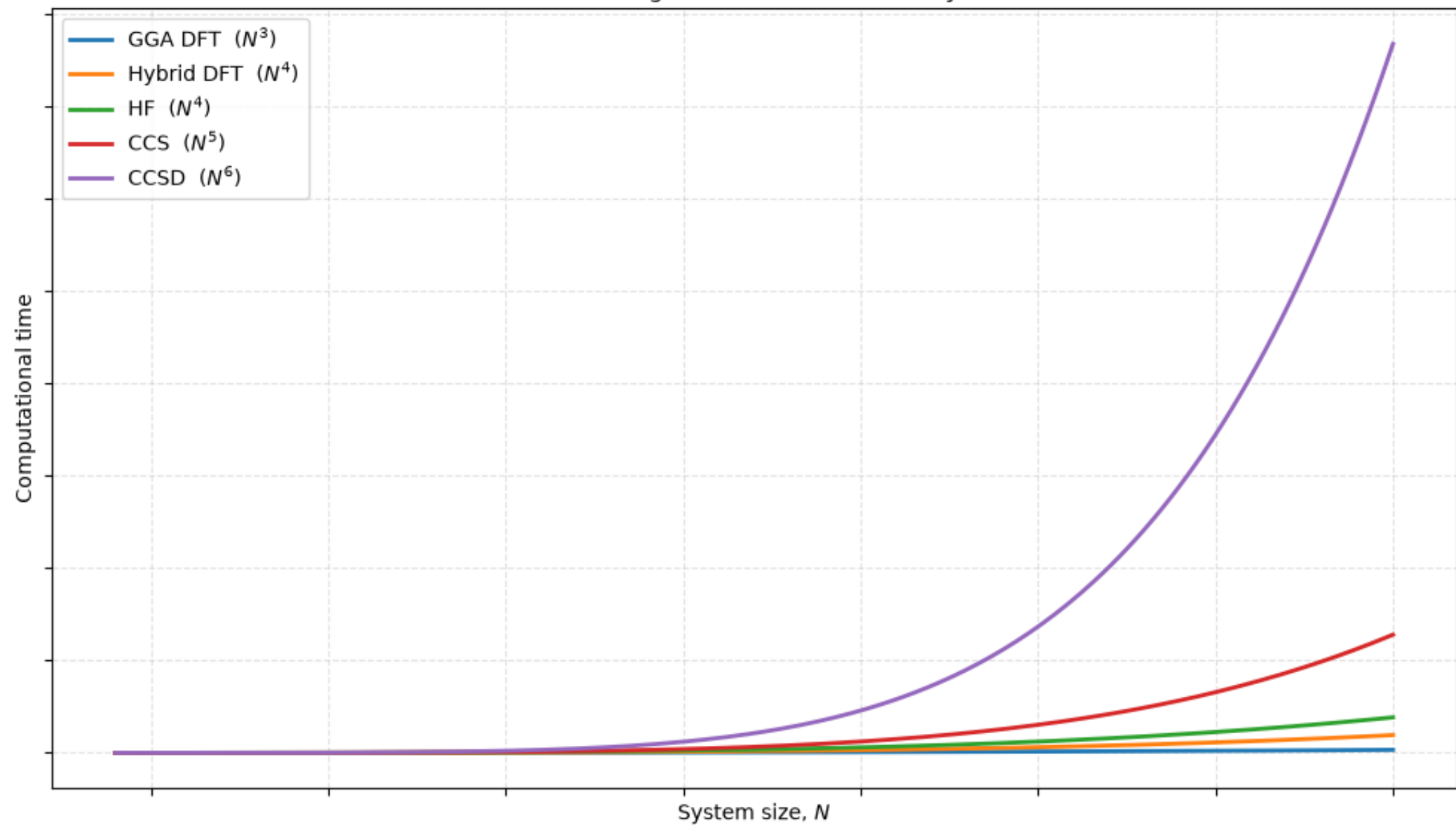


Quantum vs. Classical Calculations

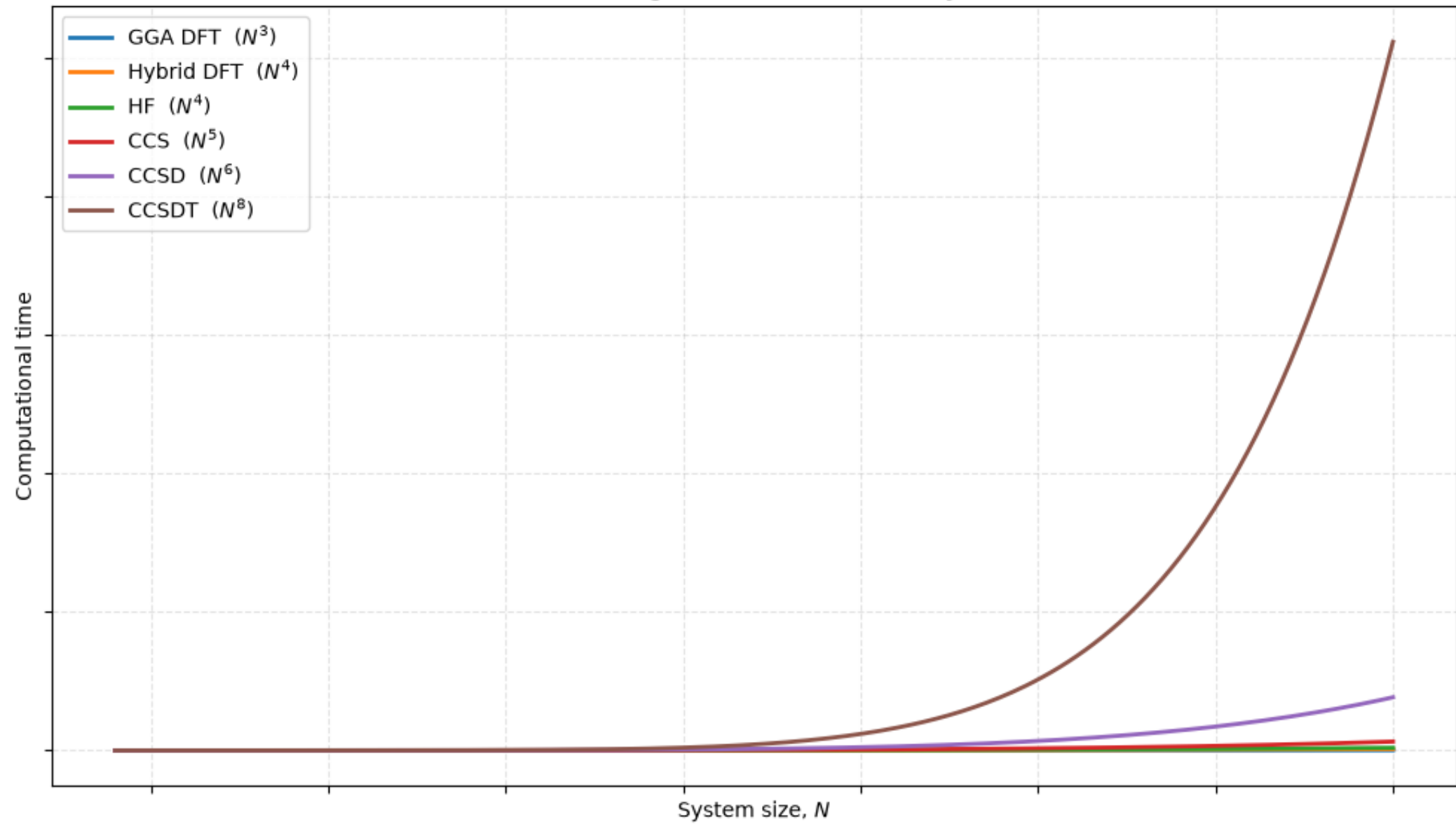
- The model you choose depends on what you want to calculate
- "All models are wrong, some are useful"
- Classical: no electronic properties, fast
- Quantum: more accurate, more properties, slow



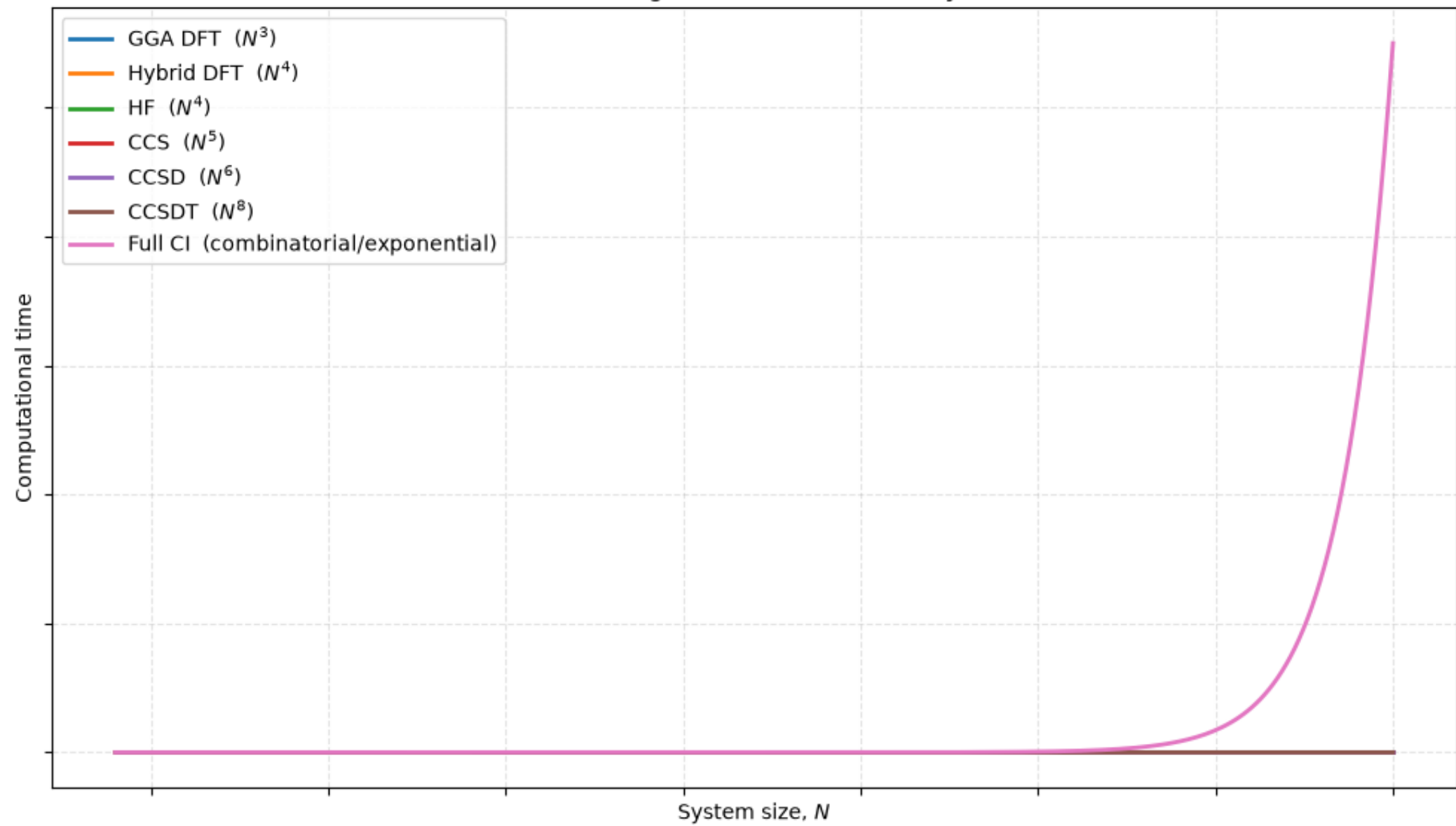
Schematic Scaling of Quantum Chemistry Methods



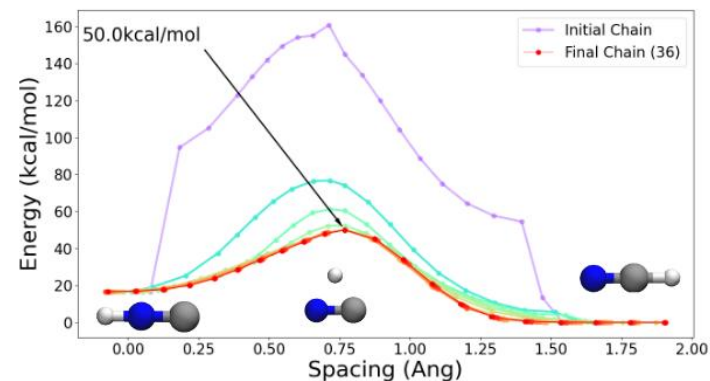
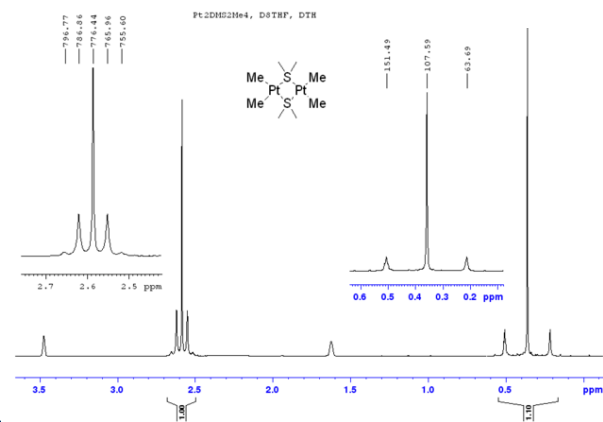
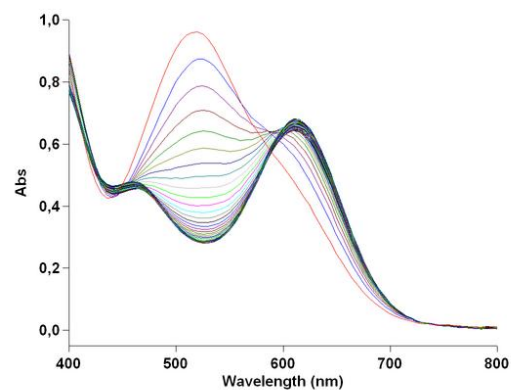
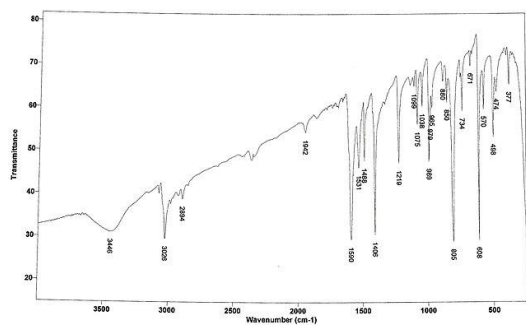
Schematic Scaling of Quantum Chemistry Methods



Schematic Scaling of Quantum Chemistry Methods

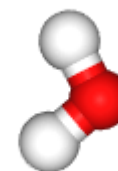


Singlepoint Calculations



- Calculation on a single fixed geometry
- Electronic properties:
 - UV/Vis, IR, CD spectra
 - NMR/Magnetic susceptibility
 - Bonding information
 - Reaction energies/Transition states

Geometry Optimization

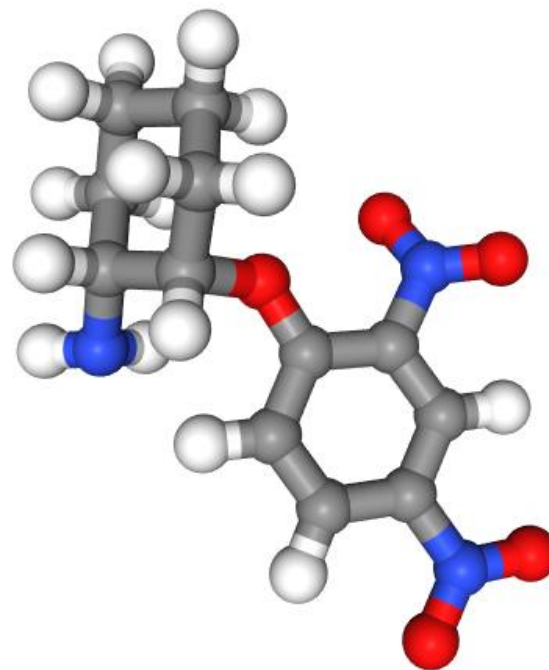


O	0.0000	0.0000	0.0000
H	0.0000	1.5875	0.0000
H	0.7938	-1.8521	0.0000

O	-0.1282	-0.1662	0.0000
H	0.3142	0.6933	0.0000
H	0.7087	-0.7918	0.0000

Why not use QM for everything?

- Geometry optimization of this molecule took **7 hours** on two cores (psi4, ω B97x/6-31G*)



QM Model Selection

- *Molecular Orbitals* (MOs) are expanded in the basis of *Atomic Orbitals* (AOs). These AOs are defined by your *basis set*
- Your *method* defines how you are calculating those MOs (what approximations)
- Combined, these are referred to as your *model*, *model chemistry*, or *level of theory*
- Often written together with a slash, with the method first

Method (Hartree-Fock)

hf/sto-3g

b3lyp/6-31g*

pbe/def2-SVP

ccsd(t)/aug-cc-pvdz

Basis set

(Slater type orbital fit with 3 gaussians)

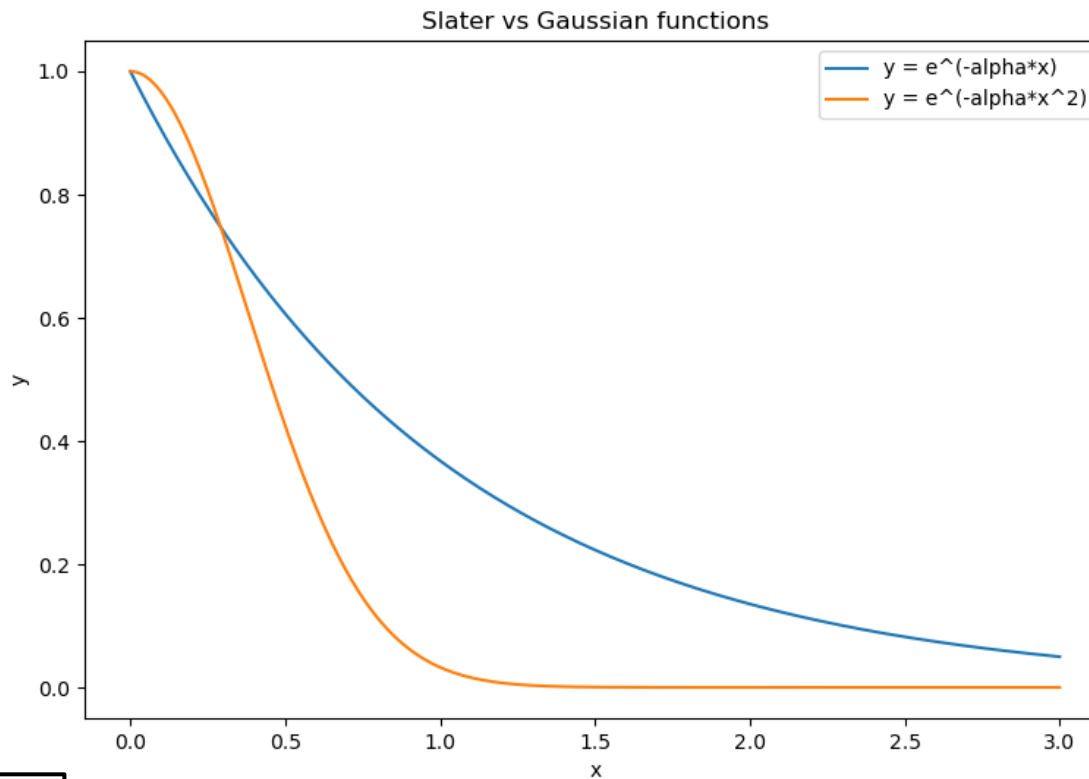
Method Hierarchy

- There is a zoo of methods, all with their strengths and weaknesses
- Hartree-Fock (HF) – very basic, but inaccurate relative to others (lacks correlation)
- Møller–Plesset perturbation theory (MP2, MPn) – Adds correlation
- Coupled Cluster (CCS, CCSD, CCSD(T)) – Very accurate, very expensive
- Density Functional Theory (DFT) – Very good balance between accuracy and cost
 - Walter Kohn – 1998 Nobel Prize in Chemistry for the development of DFT
- Composite methods (do several at once & extrapolate)
- And now machine learning!

Basis Sets

- Two common types of functions in basis sets
- Slater: $e^{-\alpha x}$
- Gaussian: $e^{-\alpha x^2}$
- Most atom centered codes use gaussian
- *In general*, larger basis sets are more accurate (but more expensive)

<https://www.basissetexchange.org>



Molecules

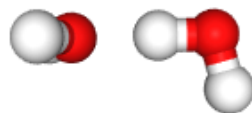
- In QM, we generally only care about:
 - Element (number of protons - Z)
 - Coordinates
 - Charge & multiplicity (unpaired electrons)
- Two units of distance
 - Angstrom ($\text{\AA}$, 10^{-10}m)
 - bohr ($5.291\,772 \times 10^{-11}\text{m}$ or $0.5291772\, \text{\AA}$)

Bonds do not exist in the QM input world

O	1.548014347149	0.00000000000000	0.060071441686
H	0.548241218221	0.00000000000000	0.038771425885
H	1.819295099755	0.00000000000000	-0.902428819751

Interaction Energy

- The main output of a singlepoint calculation is the total energy of the molecule.
- This is defined as being relative to all nuclei and all electrons being infinitely far apart. Generally, this energy is not meaningful, but *relative* energies can be useful.



- The *interaction energy* for a dimer (two molecules) is defined as $IE = E_{AB} - (E_A + E_B)$ where E_{AB} is the energy of both molecules computed together, and E_A, E_B are the energies of each of the molecules computed by themselves.